EGERTON UNIVERSITY

COMP 413: MICROPROCESSOR BASED SYSTEMS DESIGN

A REPORT FOR THE DESIGN AND IMPEMENTATION OF A MICROPROCESSOR-BASED LED CONTROL SYSTEM USING LOGIC GATES

GROUP ASSIGNMENT

MEMBERS

1. BRIAN NGUNYI – ***S13/07803/22***

2. SAMMYBRIAN NGARUIYA – ***S13/07743/22***

3. GEOFREY MACHARIA- ***S13/07772/22***

4. JOHN VUNDI- ***S13/07823/22***

5. DICKEN MURIKI- ***S13/07751/22***

MICROPROCESSOR-BASED LED CONTROL SYSTEM USING LOGIC GATES

INTRODUCTION

Microprocessor-based systems form the foundation of modern digital electronics and embedded systems. They are widely used in automation, control, and monitoring applications. This project focuses on the design and implementation of a **simple microprocessor-based LED control system** using **logic gates, switches, LEDs, resistors, memory units, and a timer**.

The system demonstrates how a microprocessor interacts with basic digital logic circuits, memory, and I/O ports to process input signals and control output devices. The project is designed for educational purposes to enhance understanding of **microprocessor architecture, digital logic concepts, and embedded system operation**.

OBJECTIVES OF THE SYSTEM

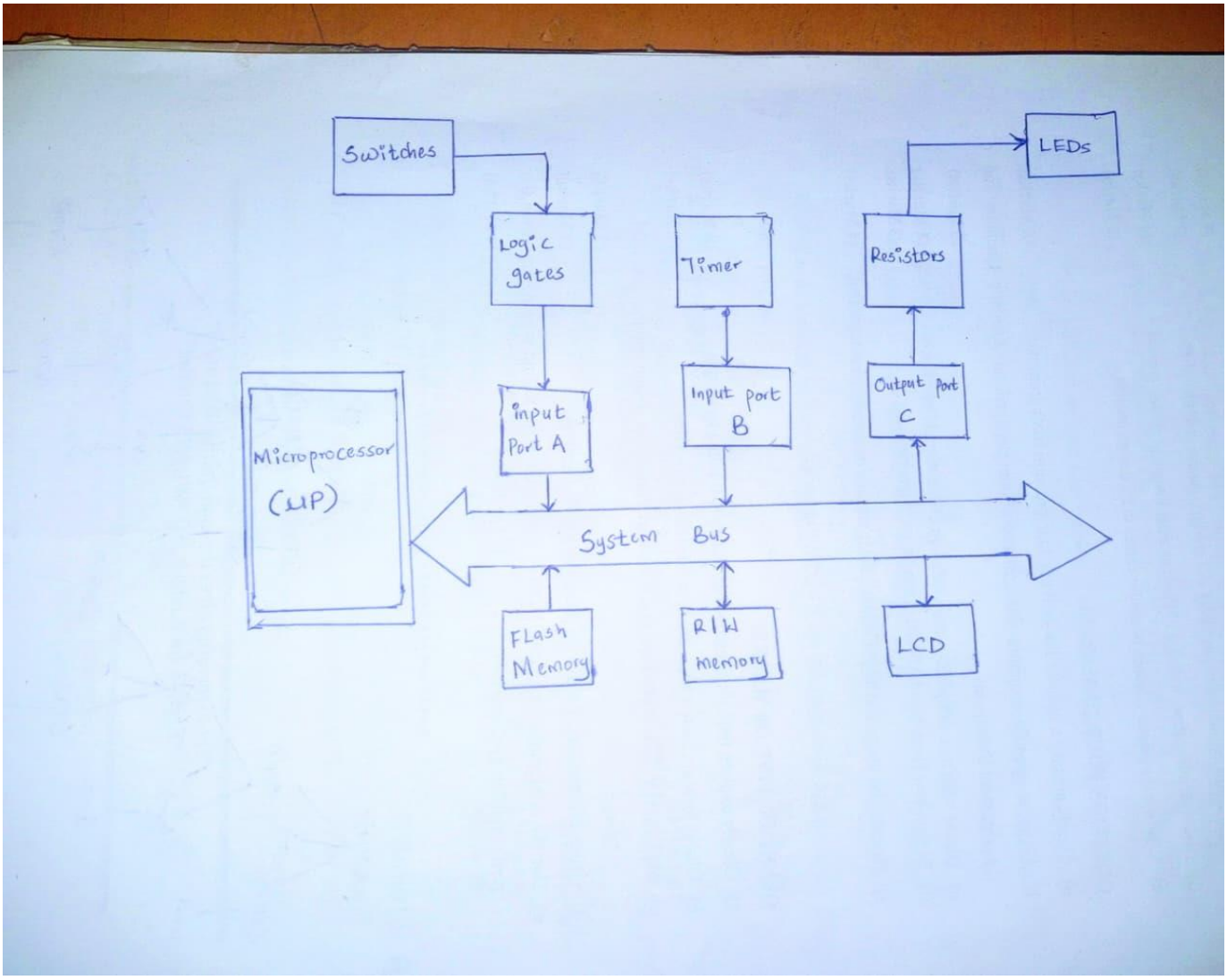
The main objectives of this system are:

- To design and implement a simple microprocessor-based control system
- To demonstrate the use of logic gates (AND, OR, NOT)
- To show how switches function as digital input devices
- To control LEDs as output indicators through output ports
- To understand the interaction between the microprocessor, memory, I/O ports, and timer
- To demonstrate the input–process–output principle in digital systems

SYSTEM OVERVIEW

The system consists of **input switches** whose signals are first processed by **logic gates**. The processed signals are then fed into the microprocessor through **input ports**. The microprocessor executes instructions stored in memory and controls output devices through **output ports**.

<u>System Block Diagram</u>



Explanation of the Diagram

1. Input Section

- **Switches** act as the **primary input devices**
- They generate digital signals (ON = 1, OFF = 0)
- These signals are first passed through **logic gates**

2. Logic Gates Block

- Logic gates (AND, OR, NOT) **process the switch inputs**
- They apply logical conditions before the signals reach the microprocessor
- This ensures only valid logic combinations are sent forward

3. Input Ports

- ***Input Port A***
    - Receives processed logic signals from the logic gates
- ***Input Port B***
    - Connected to the **Timer**
    - Allows the microprocessor to read timing or count-related signals

These ports allow external signals to enter the microprocessor system.

4. Microprocessor (CPU)

- The Microprocessor is the **control and processing unit**
- It:
    - Fetches instructions from Flash memory
    - Uses RAM for temporary data
    - Reads inputs from Ports A and B
    - Sends control signals to output ports

5. Memory Units

- ***Flash Memory (ROM)***

    - Stores the control program permanently
- ***RAM Memory***
    - Stores temporary data and variables during execution

6. Timer Block

- Provides time delays and synchronization
- Can be used for LED control, Delay control and event timing.
- Communicates with the CPU through **Input Port B**

7. Output

- ***Output Port C***
    - Sends control signals from the CPU to external devices
- ***Resistors***
    - Limit current to protect LEDs
- ***LEDs***
    - Act as visual output indicators
    - Display the result of processing

8. LCD

- Connected to the system bus
- Used for displaying system status or messages

9. System Bus

- Connects Microprocessor, memory and Input/output Ports.
    - CPU
    - Memory
    - Input/output ports

    - LCD
- Carries data, address and control signals.

System Flow

Input → Processing → Output

- **Inputs:** Switches
- **Processing:** Logic gates, microprocessor, timer, RAM, ROM
- **Outputs:** LEDs (and LCD) through current-limiting resistors

SYSTEM COMPONENTS AND EQUIPMENT

Processing Unit

- Microprocessor / Microcontroller (Raspberry Pi)

Logic Components

- AND gate IC
- OR gate IC
- NOT gate IC

Input Components

- Push-button switches
- Pull-up or pull-down resistors (10kΩ)

Output Components

- LEDs (Red, Yellow, Green)
- LED current-limiting resistors (220Ω or 330Ω)
- LCD (for optional display output)

Memory Components

- Flash memory (ROM) – stores the program

- RAM – stores temporary data and variables

Timing Component

- Timer – used for delays, synchronization, and LED blinking

Power Supply

- 5V DC power supply or USB power
- Common ground connections

Prototyping Tools

- Breadboard
- Jumper wires
- IC sockets

WORKING PRINCIPLE

1. When a switch is pressed, it generates a digital signal (HIGH or LOW).
2. This signal is first processed by the logic gates, which perform logical operations such as AND, OR, and NOT.
3. The processed logic signals are then sent to the microprocessor through Input Port A.
4. The microprocessor fetches instructions from Flash memory (ROM) and uses RAM to store temporary data during execution.
5. A timer provides timing control for operations such as LED blinking or delays.
6. Based on the program logic, the microprocessor sends output signals through Output Port C, which drive the LEDs via current-limiting resistors.
7. The LCD is used for displaying the system state.

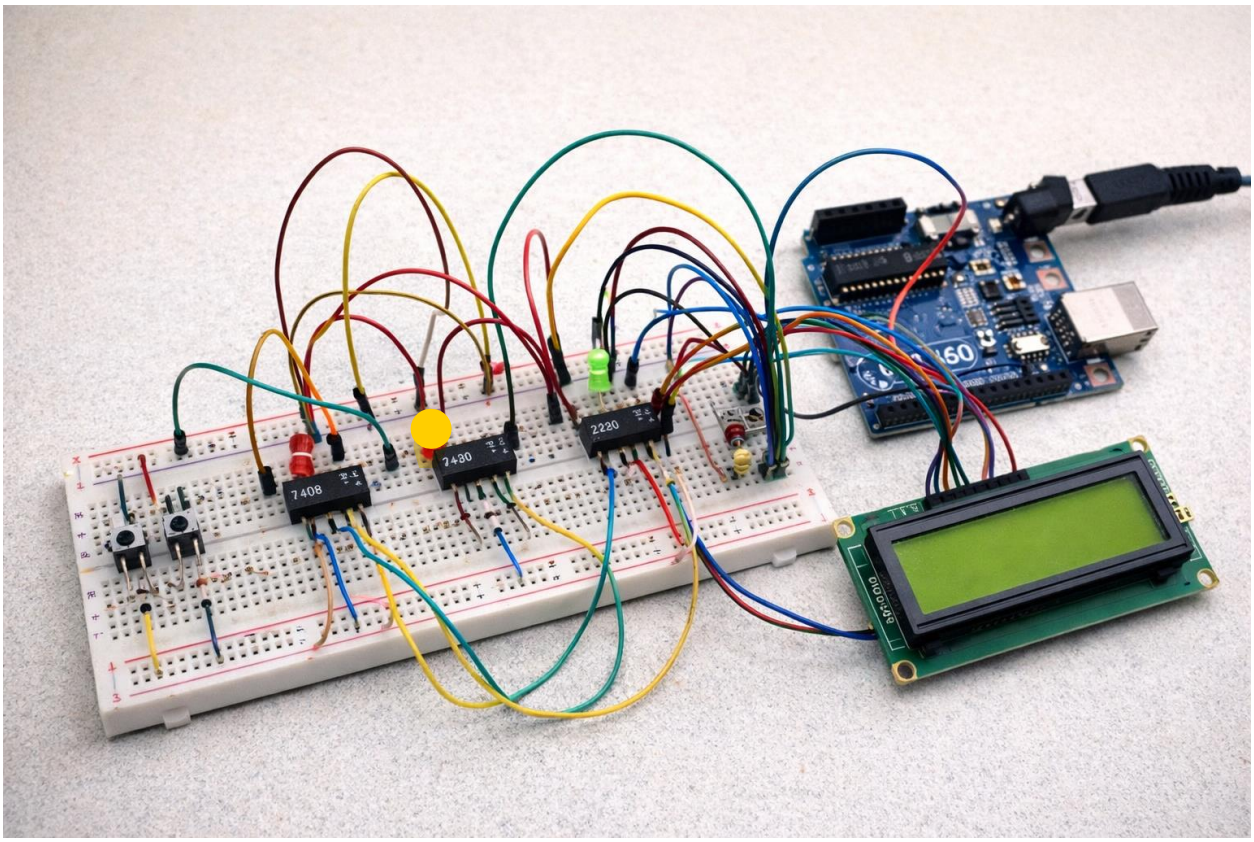
Logic Gate Output Indication

- AND gate output → Green LED ON
- OR gate output → Yellow LED ON
- NOT gate output → Red LED ON

SYSTEM DESIGN AND IMPLEMENTATION

The system was implemented on a **breadboard** to allow easy testing and modification. Logic gate ICs were placed across the breadboard's central gap, while switches, resistors, and LEDs were connected using jumper wires. The microprocessor was interfaced with input ports, output ports, memory units, and a timer through the system bus.

The microprocessor was programmed to continuously monitor input signals, process logical conditions, and update output devices in real time. Proper grounding and stable power supply connections were ensured for reliable operation.



APPLICATIONS OF THE SYSTEM

The concept demonstrated by this system can be applied in:

- Traffic light control systems
- Industrial control panels
- Digital logic and microprocessor training kits
- Alarm and security systems
- Embedded systems education and laboratory experiments

ADVANTAGES OF THE SYSTEM

- Simple and cost-effective design
- Easy to understand and implement
- Clearly demonstrates microprocessor and logic gate interaction
- Expandable to more complex control systems

LIMITATIONS OF THE SYSTEM

- Limited functionality
- Handles only basic logic operations
- No advanced communication or data storage features.

CONCLUSION

The microprocessor-based LED control system successfully demonstrates how a microprocessor works together with logic gates, memory, I/O ports, and a timer to process input signals and control output devices. The project provides a strong foundation for understanding digital systems and

embedded design. It serves as a practical and educational model that can be expanded into more advanced microprocessor-based applications.